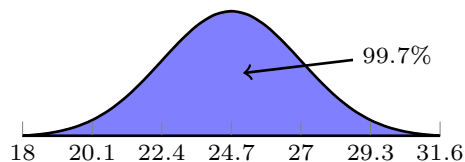
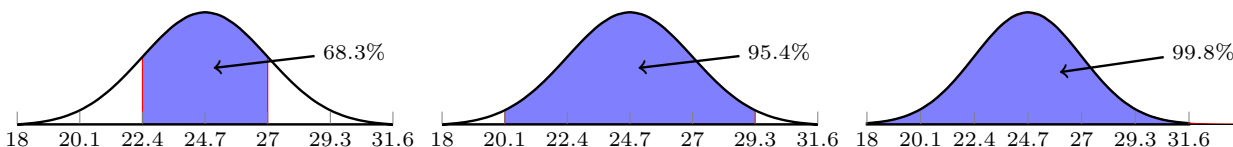


Unless stated otherwise, all problems are worth equal value and answers should be stated as decimals rounded to three decimal places. You must show your work to receive credit.

1. A population is normally distributed with mean 24.7 and standard deviation 2.3.
- a. Draw a sketch of the bell curve labeling the mean and the intervals representing 1, 2, and 3 standard deviations of the mean.



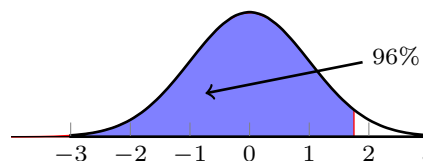
- b. Find the percentage of data lies in each of the intervals.



2. Find the following probabilities,

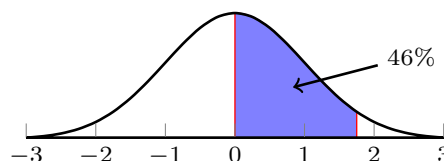
- a. $p(z < 1.75)$

$$p(z < 1.75) \approx 95.99\%$$



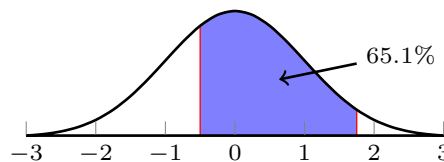
- b. $p(0 < z < 1.75)$

$$\begin{aligned} p(z < 1.75) - p(z < 0) \\ \approx 95.99\% - 50\% \approx 45.99\% \end{aligned}$$



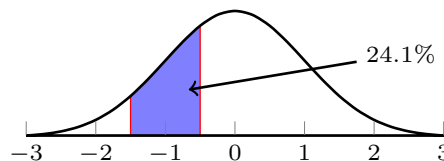
- c. $p(-.5 < z < 1.75)$

$$\begin{aligned} p(z < 1.75) - p(z < -.5) \\ \approx 95.99\% - 30.85\% \approx 65.14\% \end{aligned}$$



- d. $p(-1.5 < z < -.5)$

$$\begin{aligned} p(z < -.5) - p(z < -1.5) \\ \approx 30.85\% - 6.68\% \approx 24.17\% \end{aligned}$$

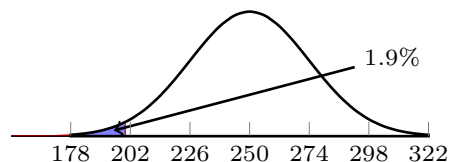


3. A population is normally distributed with a mean 250 and standard deviation 24. Find the following probabilities.

a. $p(x < 200)$

$$z = \frac{200-250}{24} = -2.08\bar{3}$$

$$p(z < -2.08) \approx 1.88\%$$



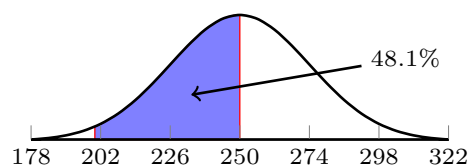
b. $p(200 < x < 250)$

$$z_{200} = \frac{200-250}{24} = -2.08\bar{3} \text{ and}$$

$$z_{250} = \frac{250-250}{24} = 0$$

$$p(-2.08 < z < 0)$$

$$\approx 50\% - 1.88\% \approx 48.12\%$$



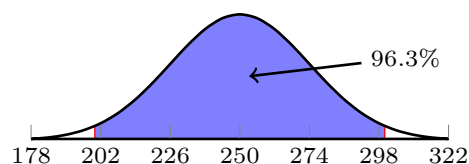
c. $p(200 < x < 300)$

$$z_{200} = \frac{200-250}{24} = -2.08\bar{3} \text{ and}$$

$$z_{300} = \frac{300-250}{24} = 2.08\bar{3}$$

$$p(-2.08 < z < 2.08)$$

$$\approx 98.12\% - 1.88\% \approx 96.24\%$$



d. $p(154 < x < 200)$

$$z_{200} = \frac{200-250}{24} = -2.08\bar{3} \text{ and}$$

$$z_{154} = \frac{154-250}{24} = -4$$

$$p(-4 < z < -2.08)$$

$$\approx 1.88\% - 0\% \approx 1.88\%$$

